Unit 3, Lesson 1: Rational Numbers and the Number Line – Part 1

Benchmark

MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.

Learning Targets

- ☐ I can demonstrate an understanding of negative numbers.
- ☐ I can recognize that a number in the real number system can have a variety of classifications.
- ☐ I can plot rational numbers on a horizontal and vertical number line.

3.1.1 Warm-Up: Creating Zero

1. Use only digits 1 to 9, at most one time each, so that the equation is true.

$$(\square + \square - \square + \square) - (\square + \square - \square + \square) = 0$$

2. Switch your answer with a partner to check. Ask your partner his/her strategy for finding his/her answer and write his/her strategy below.

3.1.2 Exploration: How Can Negative Numbers Exist?

Negative numbers are numbers less than zero.

Examples include:

- a temperature below 0 degrees
- the amount of money owed, or debt
- a contestant's score on Jeopardy (when someone misses a question, he/she loses points, so his/her score could be a negative amount)

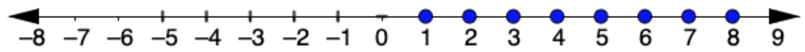
1. Discuss the examples of negative numbers with your partner.

2. Discuss with your partner which of the contexts in the table could have a value less than 0, then write an example or explain your thinking.

Context	Example or Explanation
Direction	
Points in a game	
Time	

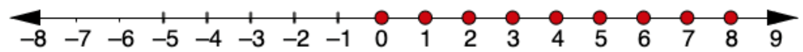
3.1.3 Guided Instruction: The Real Number System

The values in blue on the number line are some of the numbers in the set of ***natural numbers***.



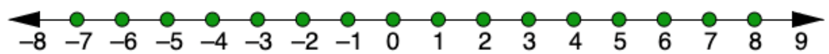
Natural numbers are the counting numbers $\{1, 2, 3, 4, 5 \ldots\}$.

The values in red on the number line are some of the numbers in the set of ***whole numbers.***



Whole numbers are natural numbers and zero. $\{0, 1, 2, 3, 4, 5 \ldots\}$

The values in green on the number line are some of the values in the set of ***integers***.



Integers are whole numbers and their opposites. $\{\ldots -4, -3, -2, -1, 0, 1, 2, 3, 4 \ldots\}$

1. Use the number lines to help to answer each of the following questions.

   a. What do you notice about the set of natural numbers and the set of whole numbers?

   b. What do you notice about the set of integers?

   c. What type of number is -10?

   d. What type of number is 27?

   e. Describe the numbers that appear in all three sets.

Camaria wondered to herself, "What about numbers such as $3\frac{1}{2}$ and -1.75?"

These numbers are examples of ***rational numbers***.

Rational numbers are numbers that can be expressed as the ratio of two integers.

2. Use the definition of rational numbers to write each as a ratio of two integers.

Number	Ratio of Two Integers
-1.5	
9	

In your study of mathematics, you will learn about ***irrational numbers,*** which are numbers that cannot be written as a ratio of two integers.

The set of ***real numbers*** is the set of all rational and irrational numbers.

3.1.4 Activity: Horizontal Number Line

Using a sheet of paper, follow the steps to make your own number line.

Step 1: Use a straightedge or a ruler to draw a horizontal line. Mark the middle point of the line and label it 0.

Step 2: To the right of 0, draw tick marks that are 1 centimeter apart. Label the tick marks $1,\ 2,\ 3 \ldots 10$.

Step 3: Fold your paper so that a vertical crease goes through 0 and the two sides of the number line match up perfectly.

Step 4: Use the fold to help you trace the tick marks that you already drew onto the opposite side of the number line. Unfold and label the tick marks $-1, -2, -3 \ldots -10$. This represents the negative side of your number line.

Use the number line to answer each of the following questions.

1. Describe how far away the number 5 is from 0.

   The number 5 *is* ______ *units to the* __________ *of* 0 *on the number line.*

2. Write a positive number and a negative number where each is further away from 0 than the number 5.

3. Describe how far away the number -2 is from 0.

   The number -2 *is* ________ *units to the* _________ *of* 0 *on the number line.*

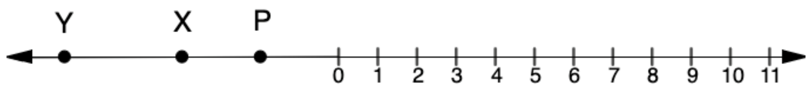
4. Write a positive number and a negative number that are each further away from 0 than -2.

3.1.5 Bring It Together: Plotting Rational Numbers on Number Lines

When plotting numbers on a number line, the sign of the number indicates which direction from 0 to move on the number line.

- On a horizontal number line, positive numbers are to the ○ right ○ left of zero.
- On a vertical number line, positive numbers are ○ above ○ below zero.
- On a horizontal number line, negative numbers are to the ○ right ○ left of zero.
- On a vertical number line, negative numbers are ○ above ○ below zero.

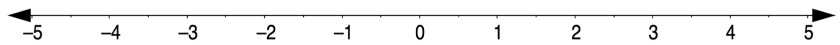
1. A number line is shown with points P, X, and Y.



a. What type of numbers are P, X, and Y?

b. Estimate the location of points P, X, and Y.

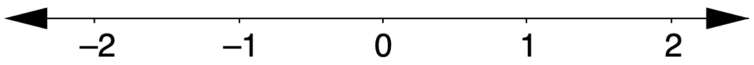
2.



a. Plot the values $-3\frac{2}{3}$ and 4.3 on the number line.

b. How did the tick marks between each integer help you determine where to plot 4.3?

c. How did the tick marks between each integer help you determine where to plot $-3\frac{2}{3}$?

3. 

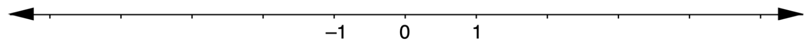
a. On the number line, make a tick mark that is halfway between each value on the number line. Label each tick mark with the value of its location.

b. Make another tick mark halfway between each value and the tick marks that were previously made. Label each with the value of its location.

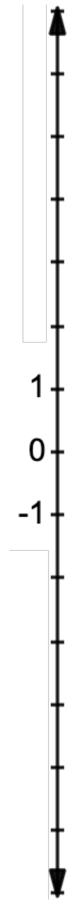
c. Plot the values $-1\frac{7}{12}$ and 0.25 on the number line.

3.1.6 Try It: Plotting Rational Numbers on a Number Line

1. Plot the numbers $-\frac{5}{2}$, -4, $\frac{1}{4}$, and 3 on the horizontal number line.



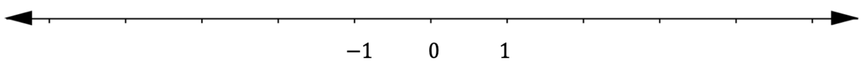
2. Plot the rational numbers -0.25, -1.8, 6.2, and 4.75 on the vertical number line.



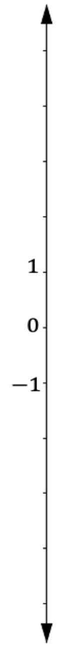
3.1.7 Your Turn!

1. Negative numbers are ○ greater ○ less than zero. On a horizontal number line, negative numbers are to the ○ right ○ left of zero. On a vertical number line, negative numbers are ○ above ○ below zero.

2. Plot the integers 2, -3, 0, -5 and 5 on the horizontal number line.

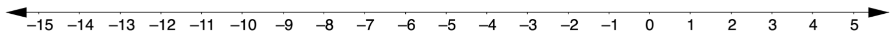


3. Plot the rational numbers $-1\frac{1}{5}$, $-\frac{4}{5}$, 2, and $\frac{6}{5}$ on the vertical number line.



3.1.8 Wrap-Up: Ticket Out the Door

1. Plot the rational numbers $\frac{-25}{4}$, $-12\frac{3}{4}$, $\frac{12}{4}$, and $\frac{1}{4}$ on the number line.



2. Explain how to estimate the value of G.